

```

        complex peak_demand[kVA];
// the peak power consumption since the last meter reading
double heatgain[Btu/h];
// the heat transferred from the enduse to the parent
double cumulative_heatgain[Btu];
// the cumulative heatgain from the enduse to the parent
double heatgain_fraction[pu];
// the fraction of the heat that goes to the parent
double current_fraction[pu];
// the fraction of total power that is constant current
double impedance_fraction[pu];
// the fraction of total power that is constant impedance
double power_fraction[pu];
// the fraction of the total power that is constant power
double power_factor; // the power factor of the load
complex constant_power[kVA];
// the constant power portion of the total load
complex constant_current[kVA];
// the constant current portion of the total load
complex constant_admittance[kVA];
// the constant admittance portion of the total load
double voltage_factor[pu];
// the voltage change factor
double breaker_amps[A]; // the rated breaker amperage
set {IS220=1} configuration; // the load configuration options
    enumeration {OFF=4294967295, NORMAL=0, ON=1} override;
    enumeration {ON=1, OFF=0, UNKNOWN=4294967295} power_
state;
}

```

```

double size[cf];
double rated_capacity[Btu/h];
double temperature[degF];
double setpoint[degF];
double deadband[degF];
timestamp next_time;
double output;
double event_temp;
double UA[Btu/degF*h];
enumeration {ON=1, OFF=0} state;
}

```

Further detailed documentation on power flow analysis is given on the website <http://gridlab-d.sourceforge.net/wiki/index.php/Powerflow>. More technical help on the software can be accessed at <https://sourceforge.net/p/gridlab-d>.

Accurate modelling of distribution networks and connected devices is possible with this open source software. The only drawback is that the original software doesn't have a GUI. It can be used as a benchmark for comparison between various systems. Results obtained can be plotted and interpreted using other open source software like Scilab, LibreOffice Calc, etc. Kindly refer to my article in the February 2017 issue of *OSFY*. As of now, Gridlabd has the flexibility for smart-grid simulation. 

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Cons

- May take some more time to mature, as OS containerisation is gradually evolving and a set of new features is getting introduced with every new OS version.
- AFW is complex to set up as it involves tie-ins with several Google consoles and EMM in use to set up the entire system. Examples of consoles are Google Admin Console, Google Play for Work, Work Profile on Android device, etc.
- AFW can be costly as it is powered by Google, which will levy charges on a per-user, per-month basis. These Google charges will be additional costs if your organisation is already paying and using some EMM tool.
- For AFW, GCDS or Google Cloud Directory (earlier known as GADS) may require to be set up to sync your organisation's Active Directory user/group with it. This may not be acceptable as per the policies of many enterprises.

OS containerisation empowers IT admins with the following key mobile management and security capabilities for BYOD:

1. Securing data on the move via app tunnel/per app VPN.
2. Securing data at rest via complete encryption.
3. Mobile app level DLP (data leak protection) by disabling screen capture, disabling copy/paste, selective app wipe, and pin protection for business app access.
4. Single sign-on.

I strongly recommend the OS containerisation integration methodology as it is provisioned by mobile OS vendors Apple and Google and has become the standard, thanks to the participation of popular EMM vendors, ISVs, enterprises, etc. It enables coverage of a wide set of mobile applications in a standardised manner without getting locked with any single EMM vendor solution, and that too in a standardised manner. 

By: Gaurav Vohra

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